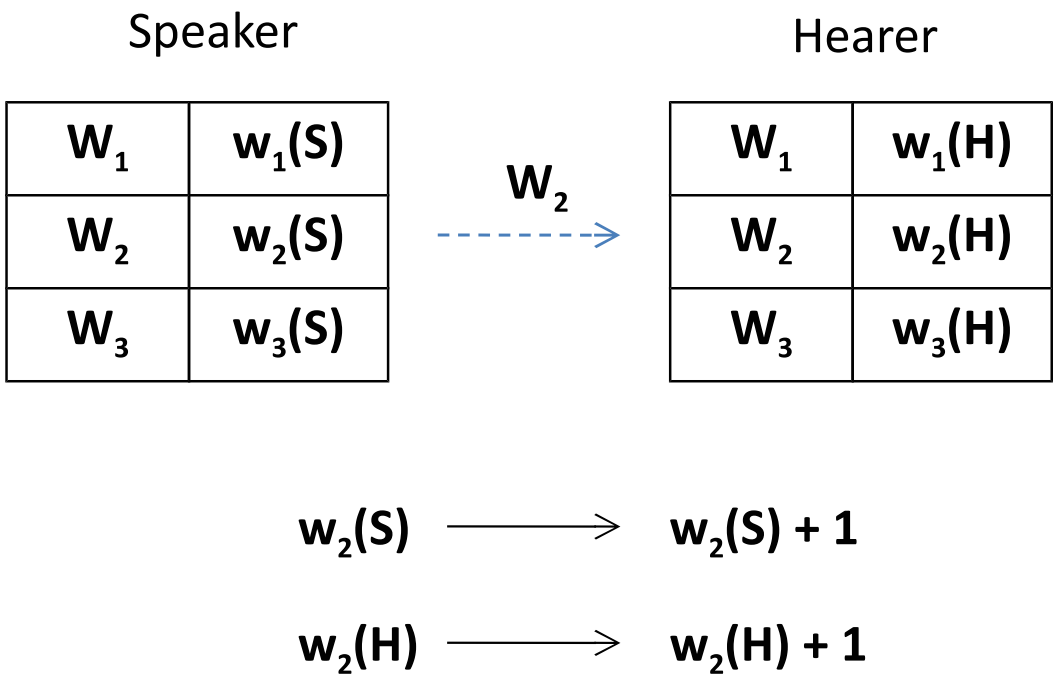

Fig 1. An elementary step of a single-object version of the model ($N_w = 3$). Using the probabilities defined in Eq (1), the speaker selects one of its words (here: W_2). Next both the speaker and the hearer increase their weights of the selected word by 1.

Provided that the object recognized by the hearer is the same as that chosen by the speaker, both agents increase the corresponding weights by 1. An elementary step of this version of the model is illustrated in Fig 2.

The above specified rules are consequences of a number of simplifying assumptions and certainly more realistic versions might be considered. For example, one might assume that the words in agents' inventories are not necessarily identical and agents could learn new words from each other. Most likely such a change would require a more sophisticated recognition mechanism and perhaps a notion of a distance between words would have to be used. Further analysis of such a version, although it seems more realistic and potentially interesting, is left for the future.

Population renewal

We also introduce a simple modification of our model (both in its single and multi- object versions), which takes into account a population renewal. The modification seems to be plausible, especially for modeling a formation of a communication system in a population of humans. In such population, when considered at a timescale of, say, hundreds of years, we should take into account a generational turnover (and possibly migrations [24]). A child learns the language of its parents but it might also acquire a (possibly different) language of its neighbors. Certainly, for a young person this is more likely to happen than for an adult. Let us notice that in urn models, due to the accumulation of weights after a large number of iterations, it is almost impossible to shift their balance (i.e., change the language). To allow for such a shift, we introduce a population renewal: With (usually small) probability p, the agent selected to be a speaker is replaced with a new agent (with all weights equal to 1), while with probability $1 - p$, the speaker acts as previously defined.

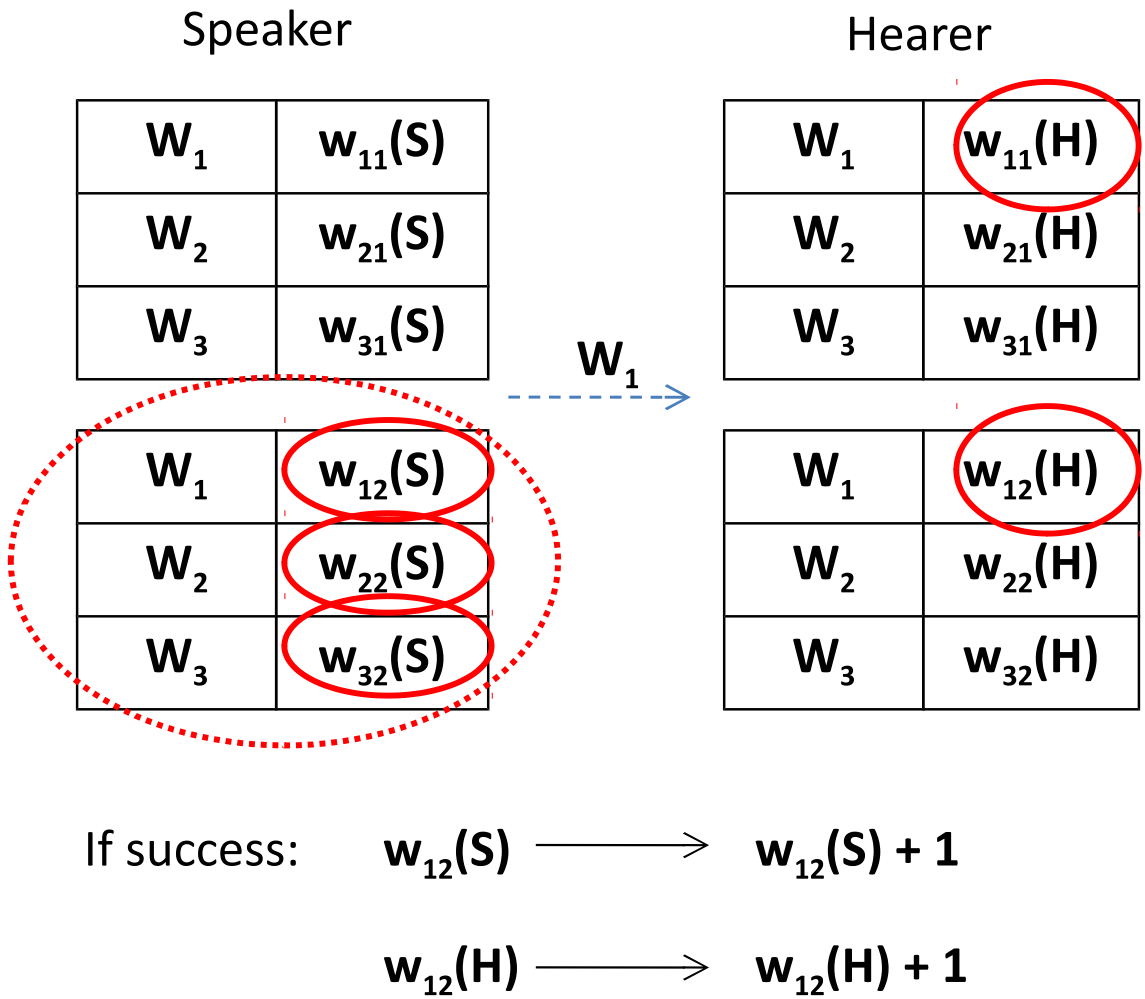

Fig 2. An elementary step of a multi-object version of the model ($N_w = 3$, $N_o = 2$). With a uniform probability $1/N_o$, the speaker chooses an object (the corresponding section of the inventory is encircled by the dotted line). Using the relevant weights (in solid circles), the speaker calculates the probabilities defined in Eq (2) and selects one of its words (here: W_1). Next the hearer tries to guess the object the speaker is talking about by calculating the probabilites (3) based on its weights of the communicated word (in circles). When the hearer's guess is correct, both agents increase their corresponding weights by 1.

Results

Single-object version

We analysed the behavior of our model for several interaction networks, namely Cartesian lattices, complete graphs and random graphs. The results obtained indicate that on Cartesian lattices the model gets stucked in a disordered structure, where consensus is only local (Fig 3). Most of the agents reach a monopolistic regime and communicate with their neighbors with the same words. There is only a small fraction of interfacial agents, which persist in a more symmetric state. For $N_w = 2$, it is tempting to confront our results with some other statistical-mechanics models. In particular, the snapshot configurations in Fig 3 suggest that initally our model coarsens, similarly to the Naming Game

and low-temperature Ising models. However, contrary to these models, the evolution of our model gets trapped in a disordered state much before reaching the uniform (mono-word) state.

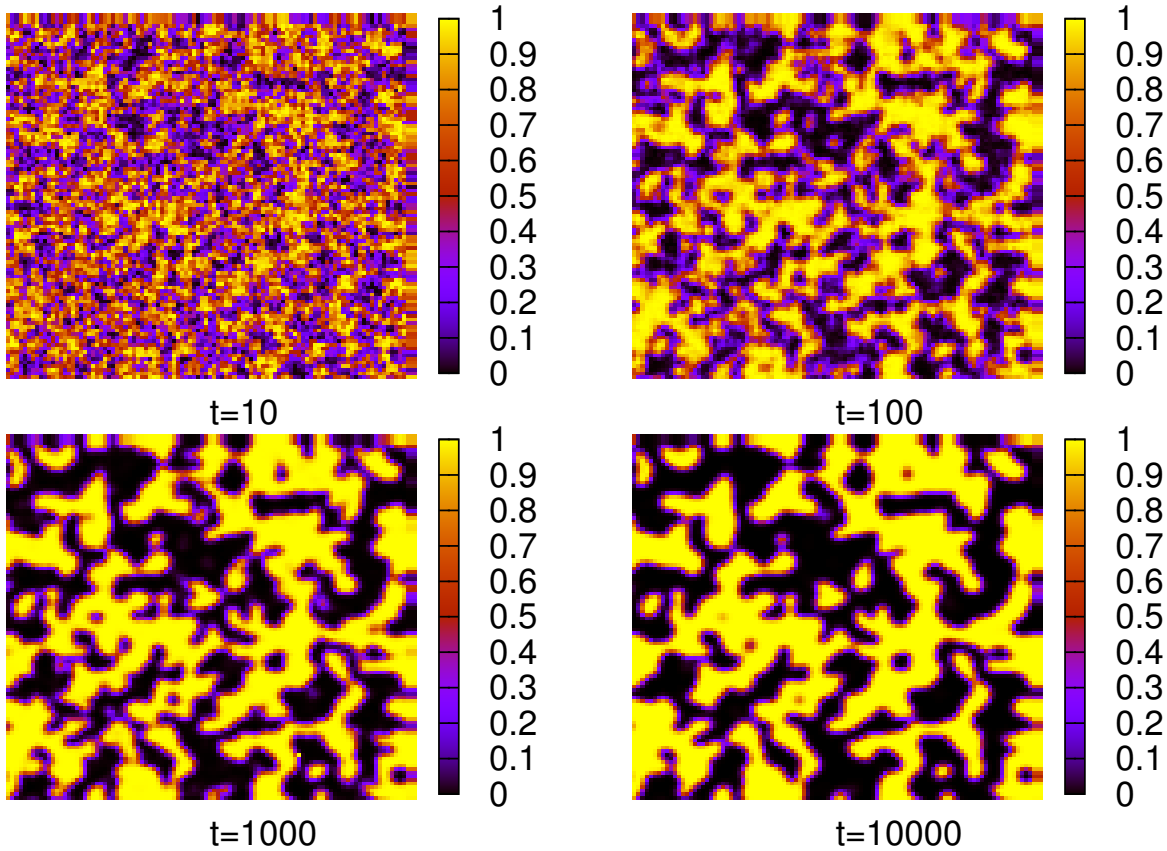

Fig 3. Spatial distribution of s_1, the probability that an agent will select word W_1 (Eq (1)). Results for a single-object model on a square lattice with $N = 10^2 \cdot 10^2 = 10^4$, $\alpha = 2$, $N_w = 2$. The dynamics traps the model in a disordered state (the configurations for, e.g., $t = 10^3$ and $t = 10^4$ differ only slightly). Since s_1 is generally close to unity or to zero, it means that almost every agent developed a strong preference toward one of the words.

To examine in more detail the behavior of the model, we calculated for $N_w = 2$ the quantities m_G and m_L defined as

$$m_G = |\frac{1}{N}\sum_A (s_1(A) - s_2(A))|,$$
$$m_L = \frac{1}{N}\sum_A |s_1(A) - s_2(A)| \tag{4}$$

where summation is over all agents A in our model.

The quantities m_G and m_L allow us to examine the global (m_G) and local (m_L) symmetries of the model. We do not present the results for $\alpha < 1$, in which case the model remains symmetric (for each agent, $s_1 = s_2 = 1/2$), and consequently, $m_G = m_L = 0$. For $\alpha > 1$, however, the asymmetry in an agent's inventory implies that $s_1 \neq s_2$ and thus $m_L > 0$. When the system is disordered, as in Fig 3, then there is no global preference toward any of the two words and $m_G = 0$.

Numerical results for $\alpha = 2$ support such an analysis. In two dimensions, relatively large m_L (Fig 4) indicates that most of the agents operate in a monopolistic regime. Moreover, m_G remains close to zero,